Benyamin Gidanian

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Summary

Applied AI/ML engineer who builds and ships systems end-to-end — from fine-tuning and evaluating models to deploying them as live, production services. Comfortable across the full stack: PyTorch model development, Python backend services (FastAPI, Django), Docker containerization, CI/CD, and cloud deployment (AWS ECS Fargate). Background spans NLP (transformer fine-tuning, token classification), computer vision (transfer learning, class imbalance), and full-stack web development. Moves fast, experiments with new tools and approaches, and focuses on shipping things that work.

Technical Skills

Core ML/AI: PyTorch, PyTorch Lightning, TensorFlow, Keras, CUDA, Torchvision, BERT, Attention Mechanisms, Hugging Face Transformers, timm, ConvNeXt, MLflow
Algorithms: Supervised & Unsupervised Learning, CNNs, Neural Networks, Decision Trees, XGBoost, Recommender Systems, Reinforcement Learning (DQN)
ML Engineering: FastAPI, Docker, RESTful APIs, CI/CD (GitHub Actions), Transfer Learning, Hyperparameter Tuning, Model Evaluation (ROC/AUC, Precision, Recall, F1)
Cloud/Infra: AWS ECS Fargate, ECR, IAM, SSM Parameter Store, Cloudflare (DNS/TLS)
Full-Stack / Web: Django, React, REST API design, SQL (SQLite)
Data: NumPy, Pandas, Feature Engineering, Data Preprocessing
Languages: Python, SQL, JavaScript, C
Tools: Git, Linux, Bash, Pytest, Ruff, Pyright, uv, Jupyter Notebooks, HuggingFace Hub

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Projects

PII Redaction

PyTorch · DeBERTa-v3 · HuggingFace Trainer/Hub · FastAPI · Docker · AWS ECS Fargate · GitHub Actions CI/CD
GitHub · Live Demo

Models: base · small · xsmall

- Fine-tuned three DeBERTa-v3 models on `ai4privacy/pii-masking-300k` for PII named entity recognition across 27 entity types (55 BIO labels: names, emails, SSNs, phone numbers, etc.), achieving macro F1 of **0.9557 (base)**, **0.9517 (small)**, and **0.9424 (xsmall)** on the held-out test set.
- Built a custom `WeightedTokenClassificationTrainer` with class-balanced cross-entropy loss to handle severe O-token imbalance; trained in two phases (frozen backbone → full fine-tune) using bf16 mixed precision, with W&B logging including per-entity F1 and interactive HTML prediction tables.
- Implemented a `PIIRedactor` class with overlapping token chunking (paragraph/sentence-aware boundary alignment), overlap resolution by max confidence score, and structured Pydantic `Entity` / `RedactionResponse` schemas; exposed via a CLI supporting `--text`, `--file`, threshold tuning, stride/max-length overrides, and JSON output.
- Built a FastAPI service with API-key-authenticated `POST /redact`, `GET /health`, `GET /model-info`, and an interactive dark-mode `GET /demo` with glassmorphism layout, drag-and-drop file upload,

and confidence badges; `POST /demo/redact` is unauthenticated but rate-limited (10/day per IP via slowapi). 14 unit tests mock the redactor across auth, rate limiting, validation, and edge cases.
- Published all three model checkpoints to HuggingFace Hub with full model cards documenting per-model F1, GPU latency benchmarks, distribution-shift behavior, and per-entity tradeoffs.
- Deployed to **AWS ECS Fargate** behind a custom domain with Cloudflare TLS termination and dynamic DNS updates on every deploy; secrets managed via SSM Parameter Store. Built a full GitHub Actions CI/CD pipeline (Ruff, Pyright, Pytest, Docker build, ECR push, ECS service update) that ships every push to `main` straight to production.

Flower Classification Inference Service

PyTorch Lightning · MLflow · FastAPI · Docker · HuggingFace Hub · GitHub Actions GitHub · **Models:** EfficientNetV2-S (serving default) · ViT-B/16 (best accuracy)

- Rebuilt a from-scratch/EfficientNet-B0 pipeline into a config-driven PyTorch Lightning system spanning 11 torchvision backbones (EfficientNet, ResNet, ViT, ConvNeXt, MobileNet, VGG) selected via a name-string registry, tracking 71 fine-tuning runs in MLflow on the 102-class Oxford Flowers dataset; best model (**ViT-B/16**) reached **0.9796 test accuracy / 0.9637 macro-F1**, with **EfficientNetV2-S** (0.9642 / 0.9364, 4x smaller, 3x faster) shipped as the serving default.
- Diagnosed and fixed a data-leakage bug that had silently built validation and test splits from the training subset, inflating all 71 runs' reported accuracy to ~0.99–1.0; re-evaluated both published models against genuinely held-out data and added a regression test (`test_splits_are_disjoint`) to prevent recurrence — a debugging exercise that also corrected the sweep's architecture rankings and checkpoint-selection logic.
- Built a custom `BackboneFinetuning` Lightning callback for two-stage fine-tuning (frozen head warm-up → full backbone unfreeze at a lower LR) and class-weighted `CrossEntropyLoss` to handle 40–258-image-per-class imbalance, tracking macro-F1 as the primary metric given the long tail.
- Restructured the FastAPI service from a flat v1 module into a v2 package (`core` / `routers` / `utils`) with an `AppState` singleton, HuggingFace Hub-backed weight loading and caching, structured logging/request middleware, and a Pydantic validation dependency enforcing file type, size, and image-dimension constraints on `POST /classify` .
- Built a GitHub Actions CI pipeline (Ruff, Pyright, Pytest, Docker build verification) and a FastAPI + Jinja training dashboard for launching runs and browsing MLflow history; containerized only the v2 API + shared `src/` for deployment.

Network — Full-Stack Social Platform

Django · React · SQLite · JavaScript GitHub

- Built a Twitter-style social platform with a Django backend (custom User, Post, Like, and Follow models, including many-to-many relations via through-models) and a React-powered frontend.
- Designed RESTful endpoints for authentication, posting, following, and liking, with server-side validation and Fetch API calls for asynchronous, no-reload post editing, liking, and pagination.

Gravitational Lens Finding — ML4SCI DeepLense (GSoC 2026 Evaluation)

PyTorch · ConvNeXt-Tiny · EfficientNet-B0 · timm GitHub

- Achieved **0.9887 AUROC** on binary gravitational lens classification (16.6:1 class imbalance) using ConvNeXt-Tiny, and **0.98 macro AUROC** on 3-class substructure classification via two-phase fine-tuning of EfficientNet-B0.

- Addressed severe class imbalance with combined `WeightedRandomSampler` oversampling and calibrated `BCEWithLogitsLoss` pos-weighting.

Additional Projects

- **ML Fundamentals Library** *(Python, NumPy, Pytest)* — Built a from-scratch ML library (Linear/Logistic Regression, K-Means, Decision Trees, multi-layer NN with manual backprop) with 100% test coverage and a custom AST-based doc generator. GitHub
- **DQN for Autonomous Lunar Landing** *(TensorFlow, Keras, Gymnasium)* — Implemented a DQN agent with experience replay and soft-update target networks, solving LunarLander-v3 with a rolling average score above 200. GitHub
- **Crossword Generator** *(Python, CSP/AC-3)* — Built a constraint satisfaction solver using AC-3 arc consistency plus MRV/LCV heuristics, reducing backtracking search space by over 80%. GitHub

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Professional Experience

One-on-One Talmud Tutor

Mesivta Bircas Yitzchok · Sept 2023 – March 2025

- Delivered individualized instruction over a 1.5-year engagement, developing structured curricula to accelerate comprehension and long-term retention.

Education & Certifications

Machine Learning Specialization — Stanford University & DeepLearning.AI (Coursera)
Supervised Learning, Neural Networks, Decision Trees, Unsupervised Learning, Recommender Systems

Harvard CS50 Suite — Harvard University (EdX)
CS50x (C, Python, SQL, Algorithms) · CS50p (OOP, Testing) · CS50ai (Search, Logic, Optimization) · CS50w (Django, React, CI/CD)